IF, IF ELSE, NESTED IF, IF ELIF ELSE,

IF Statemnet : Executes only when condition is True

IF-ELSE Statement: Chooses between 2 options Ex: Even and Odd.

IF-ELIF-ELSE Statement: Executes between multiple condition. EX: Grading System

Nested IF Statement: Checks condition inside another condtion. EX: Driving, voting

In [3]:
```
# If Stetement: Ex

# Check the number is even or odd

x = 4
r = x % 2

if r ==  0:
    print('Even Number')
```

Even Number

In [9]:
```
x = 4
r = x % 2

if r ==  0:
    print('Even Number') # Gives error as indentation
```

Even Number

In [8]:
```
# Another way to write the code as not equal to

x = 9
r = x % 2

if r ==  0:
    print('Even Number')
if r != 0:
    print('Odd Number')
```

Odd Number

In [12]:
```
# If- Else statement EX:


x = 8
r = x % 2

if r == 0:
    print('Even number')
```

```
else:
    print('Odd number')
```

```
Even number
```

In [16]:
```
# if-elif- else Statement:


marks = 75

if marks >=90:
    print('Grade A')

elif marks >= 75:
    print('Grade B')

elif marks >= 50:
    print('Grade C')
else:
    print('Fail')
```

```
Grade B
```

In [24]:
```
# Nested-if staement:

x = 4
r = x % 2

if r == 0:
    print('Even number')

    if x > 5:
        print('Greater number')
    else:
        print('Smaller number')

else:
    print('Odd number')
```

```
Even number
Smaller number
```

In [25]:
```
age = 22
has_license = False

if age >= 18:
    if has_license:
        print("You can drive.")
    else:
        print("You need a driving license.")
else:
    print("You are underage.")
```

```
You need a driving license.
```

In [ ]:
```
# LOOPS Concept

# Loops: It executed a block of code repeatedly until the condition is met.

# 1. For Loop: It is used to iterate over the sequence.

# 2. While Loop: It executes as long as condition is met.
```

In [1]:
```
# while loop ex:

i = 1 # here i is variable
```

```
while i <= 5:
    print('Data Science')
    i = i + 1 # Increment
```

```
Data Science
Data Science
Data Science
Data Science
Data Science
```

In [5]:
```
i = 1
while i <= 5:
    print('Data Science: ', i )
    i = i+1
```

```
Data Science:  1
Data Science:  2
Data Science:  3
Data Science:  4
Data Science:  5
```

In [6]:
```
i = 5
while i >= 1:
    print('Data Science:',i)
    i = i-1 # Increament in reverse
```

```
Data Science: 5
Data Science: 4
Data Science: 3
Data Science: 2
Data Science: 1
```

In [7]:
```
i = 5         # initializing
while i>1:    # condition
    print('data science : ', i)
    i = i - 1  # increment
```

```
data science :  5
data science :  4
data science :  3
data science :  2
```

In [8]:
```
i = 1

while i<=5:
    print('data science') # when we mention end then new line will not create
    j = 1
    while j<=4:
        print('technology')
        j = j + 1

    i = i + 1
    print()

    # the output which we got is very lengty but how to make them one line lets refer to belo
```

```
data science
technology
technology
technology
technology

data science
technology
technology
technology
technology

data science
technology
technology
technology
technology

data science
technology
technology
technology
technology

data science
technology
technology
technology
technology
```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]: